



Differential Calculus Topic Test I

1. Differentiate:

i. $e^{\cos x}$

$$y = e^{\cos x}$$

$$y' = -\sin x e^{\cos x}$$

iv. $\tan 3x$

$$y'(x) = 3 \sec^2 3x$$

ii. $x \cos x$

$$\frac{d}{dx}(x \cos x) = x \times -\sin x + \cos x$$

$$= \cos x - x \sin x$$

v. $x^2 e^x$

$$f'(x) = e^x \cdot 2x + x^2 e^x$$

$$= x e^x (2 + x)$$

iii. $\log_e(x^3 - 1)$

$$\frac{d}{dx}(\log_e(x^3 - 1)) = \frac{3x^2}{x^3 - 1}$$

vi. $\ln(2x + 1)$

$$(a) y = \log_e(2x + 1)$$

$$\frac{dy}{dx} = \frac{2}{2x + 1}$$

2. Find the equation of the tangent to the curve $y = 5 \ln x$ at $x = 1$.

$$(c) y = 5 \log_e x$$

$$\text{When } x = 1, y = 5 \log_e 1 = 0$$

$$\frac{dy}{dx} = 5 \times \frac{1}{x}$$

$$A(1, 0) \quad m = 5$$

$$\therefore \text{eqn of tangent is}$$

$$y - 0 = 5(x - 1)$$

$$y = 5x - 5$$

3. Find the equation of the normal to the curve $f(x) = 6 \ln(x - 1)$ at the point $(2, 0)$.

$$f(x) = 6 \ln(x - 1)$$

$$f'(x) = \frac{6}{x - 1}$$

$$\text{grad of tangent at } (2, 0)$$

$$f'(2) = \frac{6}{2 - 1} = 6$$

$$\therefore \text{norm grad } = -\frac{1}{6}$$

$$\text{eq'n of normal } y - 0 = -\frac{1}{6}(x - 2)$$

$$y = -\frac{1}{6}(x - 2)$$

$$6y = -x + 2$$

$$x + 6y - 2 = 0$$

4. If $y = e^{4x} + e^{2x}$

i. Find $\frac{dy}{dx}$

ii. Show that $\frac{d^2y}{dx^2} - \frac{6dy}{dx} + 8y = 0$.

$$(8) (a) (i) y = e^{4x} + e^{2x}$$

$$\frac{dy}{dx} = 4e^{4x} + 2e^{2x}$$

$$(ii) \frac{d^2y}{dx^2} = 16e^{4x} + 4e^{2x}$$

$$\text{LHS} = \frac{d^2y}{dx^2} - 6 \frac{dy}{dx} + 8y$$

$$= 16e^{4x} + 4e^{2x} - 6(4e^{4x} + 2e^{2x}) + 8(e^{4x} + e^{2x})$$

$$= 16e^{4x} + 4e^{2x} - 24e^{4x} - 12e^{2x} + 8e^{4x} + 8e^{2x}$$

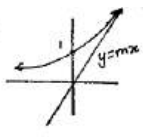
$$= 0$$

$$= \text{RHS}$$

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5. The line $y = mx$ is a tangent to the curve $y = e^{3x}$. Find the value of m .

(a)



$$y = e^{3x}$$

$$\frac{dy}{dx} = 3e^{3x}$$

gradient of $y = mx$ is m

$$\text{so } m = 3e^{3x}$$

line & curve intersect when $mx = e^{3x}$

But $m = 3e^{3x}$ so $3e^{3x}x = e^{3x}$

$$(\div \text{ by } e^{3x}) \quad 3x = 1$$

$$x = \frac{1}{3}$$

$\therefore$ intersect at $x = \frac{1}{3}$

when $x = \frac{1}{3}$ $m = 3e^{3 \times \frac{1}{3}}$

$$m = 3e$$

6. Find the equation of the tangent to the curve $y = e^{4x+1}$ at the point where $x = 0$. (Give your answer in general form.)

(a) $y = e^{4x+1}$

$$y' = 4e^{4x+1}$$

at $x = 0$

$$m_T = 4e$$

$$= 4e$$

Find y : $y = e^{4(0)+1}$

$$= e \quad (0, e)$$

Find equation:

$$y - y_1 = m(x - x_1)$$

$$y - e = 4e(x - 0)$$

$$y - e = 4ex$$

$$y - e = 4ex$$

$$4ex - y + e = 0$$

7. If $y = e^{kx}$ is a solution of the equation $\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 14y = 0$, find the value(s) of k .

$$y = e^{kx}, \frac{dy}{dx} = ke^{kx}, \frac{d^2y}{dx^2} = k^2e^{kx}$$

$$\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 14y = 0$$

$$\therefore k^2e^{kx} - 9ke^{kx} + 14e^{kx} = 0$$

$$\therefore e^{kx}(k^2 - 9k + 14) = 0$$

$$e^{kx} \neq 0, (k-2)(k-7) = 0$$

$$\therefore k = 2 \text{ or } 7$$

8. Differentiate:

i. $\frac{x}{\cos x}$

$$\frac{d}{dx} \left(\frac{x}{\cos x} \right) = \frac{\cos x \cdot 1 - x \cdot (-\sin x)}{\cos^2 x}$$

$$= \frac{\cos x + x \sin x}{\cos^2 x}$$

ii. e^{x^3}

$$\frac{d}{dx} (e^{x^3}) = 3x^2 e^{x^3}$$

iii. $x \sin x$

$$\frac{d}{dx} (x \sin x) = x \cos x + 1 \sin x$$

$$= x \cos x + \sin x$$

iv. $\log_4 x$

$$\log_4 x = \frac{\log_e x}{\log_e 4}$$

$$y' = \frac{1}{\ln 4} \times \frac{1}{x}$$

$$y' = \frac{1}{x \ln 4}$$

v. 4^x

$$4^x = e^{x \ln 4}$$

$$= e^{x \ln 4}$$

$$y' = (\ln 4) e^{x \ln 4}$$

$$y' = (\ln 4) 4^x$$

vi. $\log_{10} x^2$

$$\log_{10} x^2 = 2 \log_{10} x$$

$$= \frac{2 \ln x}{\ln 10}$$

$$y' = \frac{2}{\ln 10} \times \frac{1}{x}$$

$$y' = \frac{2}{x \ln 10}$$

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9. Differentiate:

i. $\sin^2 2x$

$$\frac{d}{dx} \sin^2 2x = \frac{d}{dx} (\sin 2x)^2$$

$$= 2 \sin 2x \cos 2x \times 2$$

$$= 4 \sin 2x \cos 2x \quad (2)$$

iv. $x^2 \cos x$

$$\frac{d}{dx} x^2 \cos x = \cos x \times 2x + x^2 \times -\sin x$$

$$= 2x \cos x - x^2 \sin x \quad (2)$$

ii. $\frac{e^x}{\log_e 2x}$

$$a) i) \frac{d}{dx} \frac{e^x}{\log_e 2x}$$

$$= \frac{(\log_e 2x)(e^x) - e^x \cdot \frac{1}{2x} \times 2}{(\log_e 2x)^2}$$

$$= \frac{e^x \log_e 2x - \frac{e^x}{x}}{(\log_e 2x)^2}$$

$$= \frac{e^x (x \log_e 2x - 1)}{x (\log_e 2x)^2} \quad (2)$$

v. $x \sin 2x$

$$a) i) \frac{d}{dx} x \sin 2x$$

$$= x \cdot 2 \cos 2x + \sin 2x$$

$$= 2x \cos 2x + \sin 2x$$

iii. $\log[x^4(x-1)^2]$

$$ii) \frac{d}{dx} \log[x^4(x-1)^2]$$

$$= \frac{d}{dx} (\log x^4 + \log (x-1)^2)$$

$$= \frac{d}{dx} (4 \log x + 2 \log (x-1))$$

$$= \frac{4}{x} + \frac{2}{x-1} \quad (2)$$

vi. $\cos x^2$

$$\frac{d}{dx} \cos x^2 = -2x \sin x^2$$

10. Find, in general form, the equation of the tangent to the curve $y = x \ln x$ at the x intercept.

$$a) y = x \ln x$$

$$y' = x \cdot \frac{1}{x} + \ln x$$

$$= 1 + \ln x$$

x int: $y = 0 \therefore x \ln x = 0$

$x = 0$ or $\log_e x = 0$

$x = 0$ not valid $\therefore e^0 = x$

valid $x = 1$

$y = 0$

$(1, 0)$

grad of tangent $= 1 + \ln 1$

$$= 1$$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 1(x - 1)$$

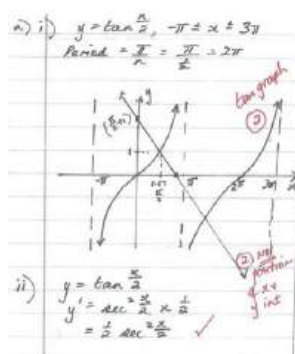
$$y = x - 1$$

$$x - y - 1 = 0 \quad (3)$$

11. i. Draw the graph of $y = \tan \frac{x}{2}$ for $-\pi \leq x \leq 3\pi$.

ii. Find the equation of the normal to the graph $y = \tan \frac{x}{2}$ when $x = \frac{\pi}{2}$.

iii. As accurately as possible, sketch this normal on to the same set of axes as the graph in (i), showing the co-ordinates of the points of intersection with the axes.



Sub $x = \frac{\pi}{2}$

$$y' = \frac{1}{2} (\sec \frac{\pi}{4})^2$$

$$= \frac{1}{2} (\sqrt{2})^2$$

$$= 1$$

$\therefore$ gradient of normal is -1 ✓

$x = \frac{\pi}{2}$ $y = \tan \frac{\pi}{4}$

$$= 1$$

$$y - 1 = -1(x - \frac{\pi}{2})$$

$$y - 1 = -x + \frac{\pi}{2}$$

$$x + y - 1 - \frac{\pi}{2} = 0$$

is eqn of normal ✓

x int: $1 + \frac{\pi}{2} \div 2.57$

y int: $1 + \frac{\pi}{2} \div 2.57 \quad (3)$



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12. Differentiate:

i. $\frac{\ln 2x}{x}$

$$\begin{aligned} \text{ii) } \frac{d}{dx} \frac{\ln 2x}{x} &= \frac{x \cdot \frac{1}{2x} \cdot 2 - \ln 2x}{x^2} \\ &= \frac{1 - \ln 2x}{x^2} \end{aligned}$$

iv. $3 \tan(4x - \frac{\pi}{4})$

$$\frac{d}{dx} 3 \tan(4x - \frac{\pi}{4}) = 12 \sec^2(4x - \frac{\pi}{4})$$

ii. $3 \tan 2x$

$$\begin{aligned} \frac{d}{dx} (3 \tan 2x) &= 3 \cdot 2 \sec^2 2x \\ &= 6 \sec^2 2x \end{aligned}$$

v. 10^{-x}

$$\begin{aligned} 10^{-x} &= e^{\ln 10^{-x}} \\ &= e^{-x \ln 10} \\ y' &= -\ln 10 \times e^{-x \ln 10} \end{aligned}$$

$$y' = -\ln 10 \times 10^{-x}$$

iii. $\ln(\sin x)$

$$\begin{aligned} \frac{d}{dx} (\ln(\sin x)) &= \frac{\cos x}{\sin x} \\ &= \cot x \end{aligned}$$

vi. $\log_5(2x^2 - 1)$

$$\log_5(2x^2 - 1) = \frac{\ln(2x^2 - 1)}{\ln 5}$$

$$y' = \frac{1}{\ln 5} \times \frac{4x}{2x^2 - 1}$$

$$y' = \frac{4x}{\ln 5 (2x^2 - 1)}$$

13. Find the equation of the tangent to the curve $y = \ln x - 2$ at the point where $x = e$. Give your answer in general form.

$$\begin{aligned} \text{c) } y &= \ln x - 2 \\ y' &= \frac{1}{x} \\ x = e, y' &= \frac{1}{e}, y = \ln e - 2 \\ &= 1 - 2 \\ &= -1 \\ \therefore y + 1 &= \frac{1}{e}(x - e) \\ ey + e &= x - e \\ \therefore x - ey - 2e &= 0 \end{aligned}$$

14. Differentiate:

i. $e^{2x} \cos x$

$$\begin{aligned} \frac{d}{dx} e^{2x} \cos x &= \cos x (2e^{2x}) + e^{2x} (-\sin x) \\ &= e^{2x} (2 \cos x - \sin x) \end{aligned}$$

ii. $\frac{\log x}{x}$

$$\frac{d}{dx} \frac{\log x}{x} = \frac{x(\frac{1}{x}) - \log x}{x^2} = \frac{1 - \log x}{x^2}$$

iii. $\frac{1}{x} + (\log x)^4$

$$\begin{aligned} \frac{d}{dx} \frac{1}{x} + (\log x)^4 &= -\frac{1}{x^2} + 4(\log x)^3 \times \frac{1}{x} \\ &= -\frac{1}{x^2} + \frac{4}{x} (\log x)^3 \end{aligned}$$

iv. $\frac{3x}{\ln x}$

$$\begin{aligned} \frac{dy}{dx} &= \frac{\ln x \times 3 - 3x \times \frac{1}{x}}{(\ln x)^2} \\ &= \frac{3 \ln x - 3}{(\ln x)^2} \end{aligned}$$



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15. The equation of the tangent to the curve $f(x) = 2e^x(x^2 + 1)$ at the point $(0, 2)$ is given by $y = ax + b$. Evaluate a and b .

$$\begin{aligned} \text{b.c) } f(x) &= 2e^x(x^2 + 1) \\ f'(x) &= 2e^x(x^2 + 1) + 2e^x \times 2x \\ &= 2e^x(x^2 + 2x + 1) \\ f'(0) &= 2(1) = 2 \\ x=0 \quad f(0) &= 2(1) = 2 \\ y - 2 &= 2(x - 0) \\ y &= 2x + 2 \\ \therefore a &= 2, \quad b = 2 \end{aligned}$$

16. Evaluate:

i. $\lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \\ &= \lim_{x \rightarrow 0} \frac{\sin \theta}{\theta} \text{ where } \theta = 3x \\ &= 1 \end{aligned}$$

ii. $\lim_{x \rightarrow 0} \frac{\sin 4x}{2x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin 4x}{2x} \\ &= 2 \times \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} \\ &= 2 \times 1 = 2 \end{aligned}$$

iii. $\lim_{x \rightarrow 0} \frac{\tan 3x}{6x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan 3x}{6x} \\ &= \frac{1}{2} \times \lim_{x \rightarrow 0} \frac{\tan 3x}{3x} \\ &= \frac{1}{2} \times 1 = \frac{1}{2} \end{aligned}$$

iv. $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{3x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{3x} &= \frac{1}{9} \lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{\frac{x}{3}} \\ &= \frac{1}{9} \times 1 \\ &= \frac{1}{9} \end{aligned}$$

v. $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} &= 4 \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{(2x)^2} \\ &= 4 \times \frac{1}{2} \\ &= 2 \end{aligned}$$

vi. $\lim_{x \rightarrow 0} \frac{1 - \cos \theta}{\theta^2}$

$$\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2} = \frac{1}{2}$$

vii. $\lim_{x \rightarrow 0} \frac{\cos(\frac{\pi}{2} - x)}{x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\cos(\frac{\pi}{2} - x)}{x} &= \lim_{x \rightarrow 0} \frac{\sin x}{x} \\ &= 1 \end{aligned}$$

viii. $\lim_{x \rightarrow 0} \frac{1 - \sin(\frac{\pi}{2} - x)}{x^2}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \sin(\frac{\pi}{2} - x)}{x^2} &= \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} \\ &= \frac{1}{2} \end{aligned}$$



Differential Calculus Topic Test I

17. Differentiate:

i. $\operatorname{cosec}(x^2 - 1)$

$$f(x) = \operatorname{cosec}(x^2 - 1)$$

$$\text{Let } u = x^2 - 1.$$

$$\frac{du}{dx} = 2x \text{ and } f'(u) = -\operatorname{cosec} u \cot u$$

$$f'(x) = \frac{d(f(u))}{du} \times \frac{du}{dx}$$

$$= -\operatorname{cosec} u \cot u \times 2x$$

$$f'(x) = -2x \operatorname{cosec}(x^2 - 1) \cot(x^2 - 1)$$

ii. $\sec x + \operatorname{cosec} x$

$$f(x) = \sec x + \operatorname{cosec} x$$

$$f'(x) = \sec x \tan x - \operatorname{cosec} x \cot x$$

iii. $\cot^2 2x$

$$f(x) = \cot^2(2x)$$

$$f(x) = (\cot 2x)^2$$

$$\text{Let } u = \cot 2x \text{ so } f(u) = u^2$$

$$\frac{du}{dx} = 2 \operatorname{cosec} 2x \text{ and } f'(u) = 2u$$

$$f'(x) = \frac{d(f(u))}{du} \times \frac{du}{dx}$$

$$= 2u \times 2 \operatorname{cosec} 2x$$

$$f'(x) = 2 \cot(2x) \times 2 \operatorname{cosec}^2(2x)$$

$$f'(x) = -4 \cot(2x) \operatorname{cosec}^2(2x)$$

iv. $\sec^2 x$

$$f(x) = \sec^2 x$$

$$f(x) = (\sec x)^2$$

$$\text{Let } u = \sec x \text{ so } f(u) = u^2$$

$$\frac{du}{dx} = \sec x \tan x \text{ and } f'(u) = 2u$$

$$f'(x) = \frac{d(f(u))}{du} \times \frac{du}{dx}$$

$$= 2u \times \sec x \tan x$$

$$= 2 \sec x \times \sec x \tan x$$

$$f'(x) = 2 \tan x \sec^2 x$$

18. Differentiate:

i. $\sqrt{\sin 2x}$

$$f(x) = \sqrt{\sin 2x} = (\sin 2x)^{\frac{1}{2}}$$

$$f'(x) = \frac{1}{2}(\sin 2x)^{-\frac{1}{2}} \times 2 \cos 2x$$

$$f'(x) = \frac{\cos 2x}{\sqrt{\sin 2x}}$$

ii. $\tan(\pi - x)$

$$f(x) = \tan(\pi - x) = -\tan(x)$$

$$f'(x) = -\sec^2 x$$

iii. $\sqrt{1 - \cos x}$

$$f(x) = \sqrt{1 - \cos x}$$

$$\text{Let } u = 1 - \cos x$$

$$f(x) = \sqrt{u} = u^{\frac{1}{2}}$$

$$\frac{du}{dx} = \frac{1}{2}u^{-\frac{1}{2}}$$

$$\frac{du}{dx} = \frac{1}{2\sqrt{1 - \cos x}}$$

$$f'(x) = \frac{du}{dx} \times \frac{du}{dx}$$

$$f'(x) = \frac{1}{2\sqrt{1 - \cos x}} \times \sin x$$

$$f'(x) = \frac{\sin x}{2\sqrt{1 - \cos x}}$$

iv. $\tan \sqrt{x}$

$$f(x) = \tan \sqrt{x}$$

$$\text{Let } u = \sqrt{x}$$

$$f(x) = \tan u$$

$$\frac{du}{dx} = \sec^2 u$$

$$\frac{du}{dx} = \sec^2 \sqrt{x}$$

$$f'(x) = \frac{du}{dx} \times \frac{du}{dx}$$

$$f'(x) = \sec^2 \sqrt{x} \times \frac{1}{2}x^{-\frac{1}{2}}$$

$$f'(x) = \frac{1}{2\sqrt{x}} \sec^2 \sqrt{x}$$

19. Differentiate:

i. $e^{x \log_e x}$

$$y = e^{x \log_e x}$$

$$\text{Let } z = x \log_e x$$

$$y = e^z$$

$$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$$

$$\text{Need to find } \frac{dz}{dx}$$

$$\text{Let } u = x \text{ and } v = \log_e x$$

$$\text{So, } \frac{du}{dx} = 1 \text{ and } \frac{dv}{dx} = \frac{1}{x}$$

$$\frac{dz}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

$$\frac{dz}{dx} = \log_e x + x \times \frac{1}{x}$$

$$\frac{dz}{dx} = \log_e(x) + 1$$

$$\text{So, } \frac{dy}{dx} = e^z (\log_e(x) + 1)$$

$$\frac{dy}{dx} = (\log_e x + 1) e^{x \log_e x}$$

ii. $x^2 \log_e x^2$

$$y = x^2 \log_e(x^2)$$

$$\text{Let } u = x^2 \text{ and } v = \log_e(x^2) = 2 \log_e x$$

$$\text{So, } \frac{du}{dx} = 2x \text{ and } \frac{dv}{dx} = \frac{2}{x}$$

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

$$\frac{dy}{dx} = 2x \log_e(x^2) + x^2 \times \frac{2}{x}$$

$$\frac{dy}{dx} = 2x \log_e(x^2) + 2x$$

$$\frac{dy}{dx} = 2x(\log_e(x^2) + 1)$$

$$\frac{dy}{dx} = 2x(2 \log_e x + 1)$$

20. Differentiate:

i. $\log_e \left(\frac{e^x + 1}{e^x - 1} \right)$

$$y = \log_e \left(\frac{e^x + 1}{e^x - 1} \right)$$

$$\text{Let } z = \frac{e^x + 1}{e^x - 1}$$

$$y = \log_e z$$

$$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$$

$$\text{Need to find } \frac{dz}{dx}$$

$$\text{Let } u = e^x + 1 \text{ and } v = e^x - 1$$

$$\text{So, } \frac{du}{dx} = e^x \text{ and } \frac{dv}{dx} = e^x$$

$$\frac{dz}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$\frac{dz}{dx} = \frac{(e^x - 1)e^x - (e^x + 1)e^x}{(e^x - 1)^2}$$

$$\frac{dz}{dx} = \frac{-2e^x}{(e^x - 1)^2}$$

$$\frac{dz}{dx} = \frac{-2e^x}{(e^x - 1)^2}$$

$$\frac{dz}{dx} = \frac{-2e^x}{(e^x - 1)^2}$$

ii. $\log_e \left(\frac{e^x + x}{\sqrt{x}} \right)$

$$f(x) = \log_e \left(\frac{e^x + x}{\sqrt{x}} \right)$$

$$= \log_e(e^x + x) - \log_e(\sqrt{x})$$

$$= \log_e(e^x + x) - \frac{1}{2} \log_e x$$

$$= \frac{e^x + 1}{e^x + x} - \frac{1}{2x}$$



Differential Calculus Topic Test 2

1. Differentiate:

i. $f(x) = \ln(3-x)$

$$f(x) = \ln(3-x)$$
$$f'(x) = \frac{-1}{3-x}$$

iv. $2\cos x \sin x$

$$2\sin x(-\sin x) + 2\cos x \cos x$$
$$2\cos^2 x - 2\sin^2 x \quad \textcircled{1} \text{ either}$$

ii. $\log\left(\frac{x+1}{x-1}\right)$

$$\frac{d}{dx} \left\{ \log(x+1) - \log(x-1) \right\}$$
$$= \frac{1}{x+1} - \frac{1}{x-1} \quad \textcircled{1}$$

v. $x^2 \tan x$

$$\text{iii) } x^2 \tan x \quad \textcircled{1}$$
$$\frac{d}{dx} = x^2 \sec^2 x + \tan x \cdot 2x$$
$$= x^2 \sec^2 x + 2x \tan x \quad \textcircled{1}$$

iii. $\sin(e^x)$

$$\text{iii) } \sin(e^x)$$
$$\frac{d}{dx} = \cos e^x \cdot e^x$$
$$= e^x \cos(e^x)$$

vi. $y = \ln\left(\frac{2x+1}{3x-1}\right)$

$$y = \log_e \left(\frac{2x+1}{3x-1} \right)$$
$$= \log_e(2x+1) - \log_e(3x-1)$$
$$y' = \frac{1}{2x+1} - \frac{1}{3x-1} \quad \textcircled{1}$$
$$y' = \frac{2(3x-1) - 3(2x+1)}{(2x+1)(3x-1)}$$
$$= \frac{6x-2-6x-3}{(2x+1)(3x-1)}$$
$$= \frac{-5}{(2x+1)(3x-1)} \quad \textcircled{1}$$

2. Find the equation of the normal to the curve $y = \tan x$ at the point where $x = \frac{\pi}{4}$.

$$y = \tan x \quad y = \tan \pi/4$$
$$y' = \sec^2 x \quad = 1$$
$$= \frac{1}{\cos^2 x}$$
$$\therefore m = \left(\frac{1}{\cos \pi/4} \right)^2$$
$$= \frac{1}{(1/\sqrt{2})^2} \quad \textcircled{1}$$
$$= 2 \quad \therefore M_N = -\frac{1}{2} \quad \textcircled{1}$$
$$\therefore y - 1 = -\frac{1}{2}(x - \pi/4)$$
$$2y - 2 = -x + \pi/4 \quad \textcircled{1}$$
$$x + 2y - 2 - \pi/4 = 0$$

Differential Calculus Topic Test 2

3. Prove that the curve $y = x^2 \ln x$ is concave upwards for $x > e^{-\frac{3}{2}}$.

$$\begin{aligned}
 y &= x^2 \ln x \\
 \frac{dy}{dx} &= (\ln x)2x + x^2 \cdot \frac{1}{x} \\
 &= x[2\ln x + 1] \quad \text{① or equivalent} \\
 \frac{d^2y}{dx^2} &= (2\ln x + 1) + x \left(\frac{2}{x} \right) \\
 &= 2\ln x + 3 \quad \text{①} \\
 \text{For concave up } \frac{d^2y}{dx^2} &> 0 \\
 \therefore 2\ln x + 3 &> 0 \\
 \ln x &> -\frac{3}{2} \\
 x &> e^{-\frac{3}{2}} \quad \text{①}
 \end{aligned}$$

4. Differentiate with respect to x :

i. $\frac{\cos 3x}{x}$

$$\begin{aligned}
 u &= \cos 3x & v &= x \\
 u' &= -3 \sin 3x & v' &= 1 \quad \text{②} \\
 \frac{dy}{dx} &= \frac{-3x \sin 3x - \cos 3x}{x^2}
 \end{aligned}$$

ii. $\sin^3 x$

$$\frac{d}{dx} (\sin x)^3 = 3 \cos x \sin^2 x$$

iii. $2 \cos^3 x$

$$\begin{aligned}
 &6 \cos^2 x (-\sin x) \\
 &= -6 \sin x \cos^2 x
 \end{aligned}$$

iv. $x^2 e^x$

$$\begin{aligned}
 u &= x^2 & v &= e^x \\
 u' &= 2x & v' &= e^x \\
 y &= x^2 e^x \\
 y' &= x^2 e^x + 2x(e^x) \\
 &= x e^x (x + 2)
 \end{aligned}$$

v. $\log \left(\frac{x}{x-1} \right)$

$$\begin{aligned}
 &\frac{1}{x} [\log x - \log(x-1)] \\
 &= \frac{1}{x} - \frac{1}{x-1} \quad \text{①}
 \end{aligned}$$

vi. $(e^{2x} + 1)^2$

$$\begin{aligned}
 \frac{dy}{dx} &= 2(e^{2x} + 1) \cdot 2e^{2x} \\
 &= 4e^{2x} (e^{2x} + 1)
 \end{aligned}$$

vii. $x^2 \cos 2x$

$$\begin{aligned}
 \frac{dy}{dx} &= \cos 2x (2x) + x^2 (-2 \sin 2x) \\
 &= 2x \cos 2x - 2x^2 \sin 2x \quad \text{②}
 \end{aligned}$$

viii. $\ln \left(\frac{x-5}{x+3} \right)$

$$\begin{aligned}
 y &= \ln \left(\frac{x-5}{x+3} \right) \\
 &= \ln(x-5) - \ln(x+3) \\
 y' &= \frac{1}{x-5} - \frac{1}{x+3} \\
 &= \frac{x+3 - (x-5)}{(x-5)(x+3)} \\
 &= \frac{8}{(x-5)(x+3)}
 \end{aligned}$$

Differential Calculus Topic Test 2

5. Find the equation of the normal to the curve $y = \ln(2x + 3)$ at the point where $x = -1$.

$$\begin{aligned} \text{a) } \frac{dy}{dx} &= \frac{2}{2x+3} \quad \text{① at } x = -1 \quad y = 0 \\ M_T &= 2 \\ M_N &= -1/2 \quad \text{①} \\ \text{eq: } y - 0 &= -1/2(x + 1) \\ 2y &= -x - 1 \quad \text{①} \\ x + 2y + 1 &= 0 \end{aligned}$$

6. Evaluate:

i. $(1 + \ln x)^3$

$$\begin{aligned} y &= (1 + \log x)^3 \\ \frac{dy}{dx} &= 3(1 + \log x)^2 \cdot \frac{1}{x} \\ &= \frac{3}{x} \cdot (1 + \log x)^2 \end{aligned}$$

ii. $3x \log x$

$$\begin{aligned} \frac{dy}{dx} &= 3 \log x + \frac{1}{x} \cdot 3x \\ &= \underline{3 \log x + 3} \end{aligned}$$

iii. $\sin^2 x$

$$\frac{dy}{dx} = \underline{2 \sin x \cos x}$$

iv. $\frac{2}{e^x}$

$$\begin{aligned} y &= \frac{2}{e^x} \\ y &= 2e^{-x} \\ y' &= -2e^{-x} \\ &= \underline{\underline{-\frac{2}{e^x}}} \end{aligned}$$

v. $\tan 2x$

$$\frac{dy}{dx} = \underline{\underline{2 \sec^2 2x}}$$

vi. $\frac{\ln x}{x}$

$$\begin{aligned} y &= \frac{\log x}{x} \\ \frac{dy}{dx} &= \frac{x \cdot \frac{1}{x} - \log x}{x^2} \\ &= \underline{\underline{\frac{1 - \log x}{x^2}}} \end{aligned}$$

vii. $e^x \cos x$

$$\begin{aligned} y &= e^x \cos x \\ \frac{dy}{dx} &= -\sin x \cdot e^x + e^x \cos x \\ &= e^x(-\sin x + \cos x) \end{aligned}$$

viii. e^{x^2-9}

$$\frac{d}{dx}(e^{x^2-9}) = 2xe^{x^2-9}$$

Differential Calculus Topic Test 2

7. Find the equation of the tangent to the curve $y = \tan x$ at the point $(\frac{\pi}{4}, 1)$.

$$\begin{aligned}
 y &= \tan x \\
 y' &= \sec^2 x \\
 \text{At } (\frac{\pi}{4}, 1) \quad y' &= \frac{\sec^2 \frac{\pi}{4}}{\cos^2 \frac{\pi}{4}} \\
 &= 2 \\
 \text{Eq'n of tangent: } y - 1 &= 2(x - \frac{\pi}{4}) \\
 y &= 2x - \frac{\pi}{2} + 1
 \end{aligned}$$

8. Differentiate:

i. $\frac{\ln x}{x^2}$

iv. $x^2 e^{-x}$

$$\begin{aligned}
 y &= \frac{\log_e x}{x^2} \\
 \frac{dy}{dx} &= \frac{x^2(\frac{1}{x}) - \log_e x(2x)}{x^4} \\
 &= \frac{x - 2x \log_e x}{x^4} \\
 &= \frac{1 - 2 \log_e x}{x^3}
 \end{aligned}$$

$$2x e^{-x} - x^2 e^{-x}$$

ii. $\log_e(\tan x)$

v. $x^2 e^{x+2}$

$$\frac{dy}{dx} = \frac{\sec^2 x}{\tan x}$$

$$\begin{aligned}
 f(x) &= x^2 e^{x+2} \\
 f'(x) &= x^2 e^{x+2} + 2x e^{x+2} \\
 &= x e^{x+2} (x+2)
 \end{aligned}$$

iii. $(1 + \cos 2x)^3$

vi. $\sin(x^3)$

$$\begin{aligned}
 f(x) &= (1 + \cos 2x)^3 \\
 f'(x) &= 3(1 + \cos 2x)^2 \times -2 \sin 2x \\
 &= -6 \sin 2x (1 + \cos 2x)^2
 \end{aligned}$$

$$y' = 3x^2 \cos x^3$$

Differential Calculus Topic Test 2

9. Find the equation of the tangent to the curve $y = \cos(\pi - x)$ at the point where $x = \frac{\pi}{3}$.

$$(c) \quad y = \cos(\pi - x)$$

$$\left(\frac{\pi}{3}, -\frac{1}{2}\right) \checkmark$$

$$\frac{dy}{dx} = \sin(\pi - x) \checkmark$$

$$\text{at } x = \frac{\pi}{3}, \text{ gradient} = \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y + \frac{1}{2} = \frac{\sqrt{3}}{2} \left(x - \frac{\pi}{3}\right) \checkmark$$

$$2y + 1 = \sqrt{3}x - \frac{\sqrt{3}\pi}{3}$$

$$\sqrt{3}x - 2y - 1 - \frac{\sqrt{3}\pi}{3} = 0$$

10. Suppose that $y = e^{kx}$.

i. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

ii. Find the value of k such that $y = 2\frac{dy}{dx} - \frac{d^2y}{dx^2}$.

$$(i) \quad y = e^{kx}$$

$$y' = ke^{kx}$$

$$y'' = k^2 e^{kx}$$

$$(ii) \quad e^{kx} = 2ke^{kx} - k^2 e^{kx}$$

$$1 = 2k - k^2$$

$$\Rightarrow k^2 - 2k + 1 = 0$$

$$(k-1)^2 = 0 \quad \therefore \boxed{k=1}$$

11. Find the slope of the tangent to the curve $y = \cos\left(x + \frac{\pi}{3}\right)$ at the point $\left(0, \frac{1}{2}\right)$.

$$(d) \quad y = \cos\left(x + \frac{\pi}{3}\right)$$

$$y' = -\sin\left(x + \frac{\pi}{3}\right)$$

2

$$\text{At } x=0 \text{ grad. tang. is } -\sin \frac{\pi}{3} = -\frac{\sqrt{3}}{2}$$

12. For the function $y = e^{-\frac{x^2}{2}}$.

i. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

ii. For what values of x is the curve of $y = e^{-\frac{x^2}{2}}$ concave down?

$$\frac{dy}{dx} = -xe^{-x^2/2}$$

$$\frac{d^2y}{dx^2} = -e^{-x^2/2} - x(-xe^{-x^2/2})$$

$$= e^{-x^2/2}(x^2 - 1)$$

$$-1 < x < 1$$

Differential Calculus Topic Test 2

13. Find the equation of the tangent to the curve $y = e^{2x-3}$ at the point $x = \frac{3}{2}$.

$$\begin{aligned} \text{d) } y &= e^{2x-3} \\ y' &= 2e^{2x-3} \\ \text{m at } x = \frac{3}{2} : m &= 2e^0 \\ &= 2 \\ x = \frac{3}{2} : y &= e^0 = 1 \\ \therefore \text{Eqn of tangent: } y - 1 &= 2\left(x - \frac{3}{2}\right) \\ y - 1 &= 2x - 3 \\ y &= 2x - 2 \end{aligned}$$

14. Evaluate:

i. $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin 2x}{x} &= \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \right) \times 2 \\ &= 2 \end{aligned}$$

ii. $\lim_{x \rightarrow 0} \frac{\tan \frac{5x}{3}}{2x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan \frac{5x}{3}}{2x} &= \lim_{x \rightarrow 0} \left(\frac{\tan \frac{5x}{3}}{\frac{5x}{3}} \right) \times \frac{5}{6} \\ &= 1 \times \frac{5}{6} \\ &= \frac{5}{6} \end{aligned}$$

iii. $\lim_{x \rightarrow 0} \frac{3x}{\sin 4x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{3x}{\sin 4x} &= \lim_{x \rightarrow 0} \left(\frac{3x}{\sin 4x} \right) \times \frac{4}{4} \\ &= 1 \times \frac{3}{4} \\ &= \frac{3}{4} \end{aligned}$$

iv. $\lim_{x \rightarrow 0} \frac{2 \sin \frac{x}{2}}{x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{2 \sin \frac{x}{2}}{x} &= \lim_{x \rightarrow 0} \left(\frac{2 \sin \frac{x}{2}}{\frac{x}{2}} \right) \times \frac{1}{2} \\ &= 1 \times 2 \times \frac{1}{2} \\ &= 1 \end{aligned}$$

v. $\lim_{x \rightarrow 0} \frac{\sin(\pi+x)}{x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin(\pi+x)}{x} &= \lim_{x \rightarrow 0} \frac{-\sin x}{x} \\ &= -1 \end{aligned}$$

vi. $\lim_{x \rightarrow 0} \frac{\sin^2 x}{x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin^2 x}{x} &= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \times \sin x \\ &= 1 \times \lim_{x \rightarrow 0} \sin x \\ &= 0 \end{aligned}$$

15. Differentiate:

i. 6^x

$$\begin{aligned} y &= 6^x \\ y &= e^{\ln 6^x} \\ y &= e^{x \ln 6} \\ y' &= (\ln 6) e^{x \ln 6} \\ &= (\ln 6) e^{\ln 6^x} \\ \boxed{y' &= (\ln 6) 6^x} \end{aligned}$$

ii. 7^{-x}

$$\begin{aligned} y &= 7^{-x} \\ y &= e^{\ln 7^{-x}} \\ y &= e^{-x \ln 7} \\ y' &= -\ln 7 \times e^{-x \ln 7} \\ &= -\ln 7 \times e^{\ln 7^{-x}} \\ \boxed{y' &= -\ln 7 \times 7^{-x}} \end{aligned}$$

Differential Calculus Topic Test 2

iii. 2^{3x-4}

$$\begin{aligned} y &= 2^{3x-4} \\ y &= e^{\ln 2^{3x-4}} \\ y &= e^{(3x-4)\ln 2} \\ y' &= 3(\ln 2) e^{(3x-4)\ln 2} \\ &= 3\ln 2 \cdot e^{\ln 2^{3x-4}} \\ \boxed{y' &= (3\ln 2) 2^{3x-4}} \end{aligned}$$

iv. $x^3 10^x$

$$\begin{aligned} \text{iv). } y &= x^3 \times 10^x \\ u &= x^3 \quad v = 10^x \\ u' &= 3x^2 \quad v' = (\ln 10) 10^x \\ y' &= 3x^2 \times 10^x + x^3 \ln 10 \times 10^x \\ \boxed{y' &= x^2 \times 10^x (3 + x \ln 10)} \end{aligned}$$

16. Find the equation of the tangent to the curve $y = 5^x$ at the point (2, 25).

$$\begin{aligned} 1. \quad y &= 5^x \\ y' &= (\ln 5) 5^x \\ \text{when } x=2, \quad y' &= (\ln 5) 5^2 \\ m_T &= 25(\ln 5) \end{aligned}$$

Tangent at (2, 25) is:

$$\begin{aligned} y - 25 &= 25(\ln 5)(x - 2) \\ y - 25 &= (25\ln 5)x - 50\ln 5 \\ \therefore y &= (25\ln 5)x - 50\ln 5 + 25 \end{aligned}$$

17. At what point(s) are the tangents to $y = 10^x$ and $y = 5^x$ parallel?

$$\begin{aligned} y &= 10^x & y &= 5^x \\ y' &= (\ln 10) 10^x & y' &= (\ln 5) 5^x \end{aligned}$$

Parallel when the gradients are the same.

$$\begin{aligned} (\ln 10) 10^x &= (\ln 5) 5^x \\ \frac{10^x}{5^x} &= \frac{\ln 5}{\ln 10} \\ \ln 2^x &= \ln \left(\frac{\ln 5}{\ln 10} \right) \\ x \times \ln 2 &= \ln \left(\frac{\ln 5}{\ln 10} \right) \\ \therefore \boxed{x &= \frac{\ln \left(\frac{\ln 5}{\ln 10} \right)}{\ln 2}} \end{aligned}$$

18. Given $y = \frac{100}{1+15e^{-0.5t}}$, find $\frac{dy}{dt}$.

$$\begin{aligned} 18). \quad y &= \frac{100}{1+15e^{-0.5t}} \\ y &= 100(1+15e^{-0.5t})^{-1} \\ \frac{dy}{dt} &= 100 \times -1(1+15e^{-0.5t})^{-2} \times 15 \times -0.5e^{-0.5t} \\ &= 750e^{-0.5t} (1+15e^{-0.5t})^{-2} \\ \boxed{\frac{dy}{dt} &= \frac{750e^{-0.5t}}{(1+15e^{-0.5t})^2}} \end{aligned}$$



Differential Calculus Topic Test 3

1. What is the derivative of $\sin(\ln x)$, where $x > 0$?

A. $\cos(\frac{1}{x})$

B. $\cos(\ln x)$

C. $\cos(\frac{\ln x}{x})$

D. $\frac{\cos(\ln x)}{x}$

1 (D)
$$\frac{d}{dx} \sin(\ln x) = \cos(\ln x) \times \frac{1}{x}$$
$$= \frac{\cos(\ln x)}{x}$$

2. Differentiate $\frac{e^x}{x+1}$.

2
$$\frac{d}{dx} \left(\frac{e^x}{x+1} \right) = \frac{(x+1) \times e^x - e^x \times 1}{(x+1)^2}$$
$$= \frac{xe^x + e^x - e^x}{(x+1)^2}$$
$$= \frac{xe^x}{(x+1)^2}$$

3. Differentiate $x^2 \tan x$.

$$\frac{d}{dx} (x^2 \tan x)$$
$$= x^2 \times \sec^2 x + \tan x \times (2x)$$
$$= x(x \sec^2 x + 2 \tan x)$$

4. Let $f(x) = xe^{3x}$. Evaluate $f'(0)$.

$$f(x) = xe^{3x}$$
$$f'(x) = x \times 3e^{3x} + e^{3x} \times 1$$
$$= e^{3x}(3x + 1)$$
$$f'(0) = e^0(1) = 1$$

5. Find the equation of the tangent to the curve $y = \cos 2x$ at $x = \frac{\pi}{6}$.

$$y = \cos 2x$$
$$y' = -2 \sin 2x$$

At the point $(\frac{\pi}{6}, \frac{1}{2})$

$$y' = -2 \sin (2 \times \frac{\pi}{6}) = -\sqrt{3}$$

Equation of the tangent:

$$y - y_1 = m(x - x_1)$$
$$y - \frac{1}{2} = -\sqrt{3} \left(x - \frac{\pi}{6} \right)$$
$$6y - 3 = -6\sqrt{3}x + \sqrt{3}\pi$$
$$6\sqrt{3}x + 6y - 3 - \sqrt{3}\pi = 0$$

6. Let $f(x) = \frac{(2x+1)(x+4)}{\sqrt{x}}$, $x > 0$.

- a. Show that $f(x)$ can be written in the form $Ax^{\frac{3}{2}} + Bx^{\frac{1}{2}} + Cx^{-\frac{1}{2}}$. Find the values of A , B and C .
- b. Find $f'(x)$.

6(a)

$$f(x) = \frac{(2x+1)(x+4)}{\sqrt{x}}$$
$$= \frac{2x^2 + 9x + 4}{\sqrt{x}}$$
$$= 2x^{\frac{3}{2}} + 9x^{\frac{1}{2}} + 4x^{-\frac{1}{2}}$$
$$\therefore A = 2, B = 9 \text{ and } C = 4$$

6(b)

$$f(x) = 2x^{\frac{3}{2}} + 9x^{\frac{1}{2}} + 4x^{-\frac{1}{2}}$$
$$f'(x) = 3x^{\frac{1}{2}} + \frac{9}{2}x^{-\frac{1}{2}} - 2x^{-\frac{3}{2}}$$



Differential Calculus Topic Test 3

7. Let $f(x) = \sin\left(\frac{x}{2}\right) + \tan\left(\frac{x}{2}\right) + 2$.

a. Find $f'(x)$.

b. Solve the equation $f'(x) = 0$ for $x \in [0, 2\pi]$.

7(a)

$$f(x) = \sin\left(\frac{x}{2}\right) + \tan\left(\frac{x}{2}\right) + 2$$

$$\begin{aligned} f'(x) &= \frac{1}{2} \cos\left(\frac{x}{2}\right) + \frac{1}{2} \sec^2\left(\frac{x}{2}\right) \\ &= \frac{1}{2} \left[\cos\left(\frac{x}{2}\right) + \sec^2\left(\frac{x}{2}\right) \right] \end{aligned}$$

7(b)

$$\frac{1}{2} \left[\cos\left(\frac{x}{2}\right) + \sec^2\left(\frac{x}{2}\right) \right] = 0$$

$$\cos\left(\frac{x}{2}\right) + \frac{1}{\cos^2\left(\frac{x}{2}\right)} = 0$$

$$\cos^3\left(\frac{x}{2}\right) = -1$$

$$\cos\left(\frac{x}{2}\right) = -1$$

$$\frac{x}{2} = \pi$$

$$\therefore x = 2\pi$$

8. What is the derivative of e^{x^2} ?

A. $x^2 e^{x^2-1}$

B. $2xe^{2x}$

C. $2xe^{x^2}$

D. $2e^{x^2}$

8 (C)

$$\frac{d}{dx}(e^{x^2}) = (e^{x^2}) \times 2x = 2xe^{x^2}$$

9. For $f(x) = \frac{\cos x}{2x+2}$, find $f'(x)$.

$$\begin{aligned} f(x) &= \frac{\cos x}{2x+2} \\ f'(x) &= \frac{(2x+2)(-1)\sin x - \cos x \times 2}{(2x+2)^2} \\ &= \frac{(-2x-2)\sin x - 2\cos x}{(2x+2)^2} \end{aligned}$$

10. Differentiate:

a. $e^x + 2^x$

b. $x^2 3^x$

c. $\frac{4^x}{x}$

10(a)

$$\frac{d}{dx}(e^x + 2^x) = e^x + (\ln 2) 2^x$$

10(b)

$$\begin{aligned} \frac{d}{dx}(x^2 3^x) &= x^2 \times (\ln 3) 3^x + 3^x \times 2x \\ &= x 3^x (x \ln 3 + 2) \end{aligned}$$

10(c)

$$\begin{aligned} \frac{d}{dx}\left(\frac{4^x}{x}\right) &= \frac{x \times (\ln 4) 4^x - 4^x \times 1}{x^2} \\ &= \frac{4^x (x \ln 4 - 1)}{x^2} \end{aligned}$$

11. The line $y = mx + c$ is a tangent to the curve $y = x^3 - 3x + 2$ at the point $(-2, 0)$. What are the values of m and c ?

$$y = x^3 - 3x + 2$$

$$y' = 3x^2 - 3$$

At the point $(-2, 0)$

$$y' = 3 \times (-2)^2 - 3 = 9$$

Equation of the tangent:

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 9(x - (-2))$$

$$y = 9x + 18$$

$$\therefore m = 9 \text{ and } c = 18$$



Differential Calculus Topic Test 3

12. Find the equation of the tangent to the curve $y = x^2 + 4x - 7$ at the point $(1, -2)$?

$$y = x^2 + 4x - 7$$

$$y' = 2x + 4$$

At the point $(1, -2)$

$$y' = 2 \times 1 + 4 = 6$$

Equation of the tangent:

$$y - y_1 = m(x - x_1)$$

$$y - (-2) = 6(x - 1)$$

$$y + 2 = 6x - 6$$

$$6x - y - 8 = 0$$

13. a. Show that $\frac{e^{3x+4}}{e^x} = e^{2x} + 4e^{-x}$.

b. Hence, find the derivative of the above function.

c. If $y = xe^{2x}$, prove that $\frac{dy}{dx} - 2y = e^{2x}$.

13(a)

$$\begin{aligned}\frac{e^{3x+4}}{e^x} &= \frac{e^{3x}}{e^x} + \frac{4}{e^x} \\ &= e^{2x} + 4e^{-x}\end{aligned}$$

13(b)

$$\begin{aligned}\frac{d}{dx}\left(\frac{e^{3x+4}}{e^x}\right) &= \frac{d}{dx}(e^{2x} + 4e^{-x}) \\ &= 2e^{2x} - 4e^{-x}\end{aligned}$$

13(c)

$$\begin{aligned}y &= xe^{2x} \\ \frac{dy}{dx} &= x \times 2e^{2x} + e^{2x} \times 1 \\ &= 2xe^{2x} + e^{2x} \\ \frac{dy}{dx} - 2y &= 2xe^{2x} + e^{2x} - 2(xe^{2x}) \\ &= 2xe^{2x} + e^{2x} - 2xe^{2x} \\ &= e^{2x}\end{aligned}$$

14. If x is a small angle measure in radians, show that:

a. $\cos\left(\frac{\pi}{6} - x\right) \approx \frac{1}{2}(\sqrt{3} + x)$

b. $\tan\left(\frac{\pi}{4} + x\right) \approx \frac{1+x}{1-x}$

14(a)

$$\begin{aligned}\cos\left(\frac{\pi}{6} - x\right) &= \cos\frac{\pi}{6} \times \cos x + \sin\frac{\pi}{6} \times \sin x \\ &\approx \frac{\sqrt{3}}{2} \times 1 + \frac{1}{2} \times x \\ &\approx \frac{1}{2}(\sqrt{3} + x)\end{aligned}$$

14(b)

$$\begin{aligned}\tan\left(\frac{\pi}{4} + x\right) &= \frac{\tan\frac{\pi}{4} + \tan x}{1 - \tan\frac{\pi}{4} \times \tan x} \\ &\approx \frac{1 + x}{1 - x}\end{aligned}$$

15 (B)

$$\begin{aligned}\frac{d}{dx} \ln(\cos x) &= \frac{1}{\cos x} \times -\sin x \\ &= -\tan x\end{aligned}$$

15. Find the derivative of $\ln(\cos x)$?

$$\begin{aligned}\frac{d}{dx} \ln(\cos x) &= \frac{1}{\cos x} \times -\sin x \\ &= -\tan x\end{aligned}$$



Differential Calculus Topic Test 3

16. Differentiate $\frac{x+2}{3x-4}$.

$$\begin{aligned}\frac{d}{dx}\left(\frac{x+2}{3x-4}\right) &= \frac{(3x-4)(1) - (x+2)(3)}{(3x-4)^2} \\ &= \frac{3x-4-3x-6}{(3x-4)^2} \\ &= \frac{-10}{(3x-4)^2}\end{aligned}$$

17. Differentiate $y = xe^{3x}$.

$$\begin{aligned}y &= xe^{3x} \\ \frac{dy}{dx} &= 3xe^{3x} + e^{3x} \\ &= e^{3x}(1+3x)\end{aligned}$$

18. What is the equation of the normal to the curve $y = x^2 - 4x$ at the point $(1, -3)$?

$$y = x^2 - 4x$$

$$y' = 2x - 4$$

At the point $(1, -3)$

$$y' = 2 \times 1 - 4 = -2$$

Gradient of the normal:

$$m_1 m_2 = -1$$

$$m \times -2 = -1$$

$$m = \frac{1}{2}$$

Equation of the normal:

$$y - y_1 = m(x - x_1)$$

$$y - (-3) = 0.5(x - 1)$$

$$2y + 6 = x - 1$$

$$x - 2y - 7 = 0$$

19. Differentiate:

a. $x + \log_2 x$

b. $x^3 \log_4 x$

c. $\frac{\log_a x}{x^2}$

19(a)

$$\frac{d}{dx}(x + \log_2 x) = 1 + \frac{1}{x \ln 2}$$

19(b)

$$\begin{aligned}\frac{d}{dx}(x^3 \log_4 x) &= x^3 \times \frac{1}{x \ln 4} + \log_4 x \times 3x^2 \\ &= x^2 \left(\frac{1}{\ln 4} + 3 \log_4 x \right)\end{aligned}$$

19(c)

$$\begin{aligned}\frac{d}{dx}\left(\frac{\log_a x}{x^2}\right) &= \frac{x^2 \times \frac{1}{x \ln a} - \log_a x \times 2x}{x^4} \\ &= \frac{\frac{1}{\ln a} - 2 \log_a x}{x^3}\end{aligned}$$

20. Differentiate $x \ln x$ with respect to x .

$$y = x \ln x$$

$$\begin{aligned}y' &= x \times \frac{1}{x} + \ln x \times 1 \\ &= 1 + \ln x\end{aligned}$$

21. Find the gradient of the curve $y = \tan x$ at the point where $x = \frac{\pi}{8}$.

Give your answer correct to 3 significant figures.

$$y = \tan x$$

$$y' = \sec^2 x$$

At the point where $x = \frac{\pi}{8}$

$$y' = \sec^2 \frac{\pi}{8}$$

$$= \frac{1}{\cos^2 \frac{\pi}{8}}$$

$$\approx 1.17$$

$\therefore$ Gradient of the tangent is 1.17.



Differential Calculus Topic Test 3

22. Simplify $\lim_{x \rightarrow 0} \frac{\tan \frac{x}{3}}{x}$.

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\tan \frac{x}{3}}{x} &= \frac{1}{3} \times \lim_{\frac{x}{3} \rightarrow 0} \frac{\tan \frac{x}{3}}{\frac{x}{3}} \\ &= \frac{1}{3}\end{aligned}$$

23. Find the gradient of the curve $y = \ln x$ at the point where it crosses the x - axis.

Curve crosses the x -axis when $y = 0$.

$x = e^y$. The point is $(e^y, 0)$.

$$y = \ln x$$

$$y' = \frac{1}{x}$$

At the point $(e^y, 0)$:

$$m = \frac{1}{e^y} = e^{-y}$$

24. What is the derivative of $y = 2\sin 3x - 3\tan x$ at $x = 0$?

$$y = 2\sin 3x - 3\tan x$$

$$y' = 2 \times 3\cos 3x - 3 \times \sec^2 x$$

$$= 6\cos 3x - 3\sec^2 x$$

$$\text{At } x = 0$$

$$y' = 6 - 3 = 3$$

25. Differentiate $(e^x + x)^5$.

$$y = (e^x + x)^5$$

$$y' = 5(e^x + x)^4 \times (e^x + 1)$$

$$= 5(e^x + 1)(e^x + x)^4$$

26. Differentiate $y = (x + 4)\ln x$.

$$y = (x + 4)\ln x$$

$$y' = (x + 4) \times \frac{1}{x} + \ln x \times 1$$

$$= \ln x + \frac{x + 4}{x}$$

27. Find $f'(x)$, where $f(x) = \frac{x^2 + 3}{x - 1}$.

$$f(x) = \frac{x^2 + 3}{x - 1}$$

$$f'(x) = \frac{(x - 1) \times 2x - (x^2 + 3) \times 1}{(x - 1)^2}$$

$$= \frac{2x^2 - 2x - x^2 - 3}{(x - 1)^2}$$

$$= \frac{x^2 - 2x - 3}{(x - 1)^2}$$



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28. What is the equation of the tangent to the curve $y = x^2$ at the point $x = 3$?

$$\begin{aligned}y &= x^2 \\y' &= 2x \\ \text{At the point } x &= 3 \\y' &= 2 \times 3 = 6 \\ \text{Equation of the tangent} \\y - y_1 &= m(x - x_1) \\y - 9 &= 6(x - 3) \\y - 9 &= 6x - 18 \\6x - y - 9 &= 0\end{aligned}$$

29. a. Show that the tangent at the point P where $x = 3$ on the curve $y = e^x$ has the equation $e^3x - y - 2e^3 = 0$.
b. The tangent at P meets the x -axis at Q . If R is the foot of the perpendicular from P on the x -axis, find the length of QR .

29(a)

$$\begin{aligned}y &= e^x \\y' &= e^x \\ \text{At the point } x &= 3: \\y' &= e^3 \\ \text{Equation of the tangent:} \\y - y_1 &= m(x - x_1) \\y - e^3 &= e^3(x - 3) \\y - e^3 &= e^3x - 3e^3 \\e^3x - y - 2e^3 &= 0\end{aligned}$$

29(b)

To find point Q ($y = 0$):

$$\begin{aligned}e^3x - 0 - 2e^3 &= 0 \\e^3x &= 2e^3 \\x &= 2 \\\therefore Q(2, 0) \\ \text{Coordinates } R(3, 0) \\ \therefore \text{Length of } QR &\text{ is 1 unit.}\end{aligned}$$

30. Simplify $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$.

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin 5x}{x} &= 5 \lim_{5x \rightarrow 0} \frac{\sin 5x}{5x} \\&= 5\end{aligned}$$

31. Differentiate $e^x \cos x$.

$$\begin{aligned}\frac{d}{dx} e^x \cos x \\&= e^x \times -\sin x + \cos x \times e^x \\&= e^x (\cos x - \sin x)\end{aligned}$$

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32. Find the value of k if $y = e^{kx} \sin x$ and $\frac{dy}{dx} - 3y = e^{kx} \cos x$.

$$\begin{aligned} y &= e^{kx} \sin x \\ \frac{dy}{dx} &= e^{kx} \times \cos x + \sin x \times k e^{kx} \\ &= e^{kx} (\cos x + k \sin x) \\ \frac{dy}{dx} - 3y &= e^{kx} \cos x \\ e^{kx} (\cos x + k \sin x) - 3e^{kx} \sin x &= e^{kx} \cos x \\ k e^{kx} \sin x - 3e^{kx} \sin x &= 0 \\ e^{kx} \sin x (k - 3) &= 0 \\ k &= 3 \end{aligned}$$

33. Differentiate $\frac{x^3}{x+1}$.

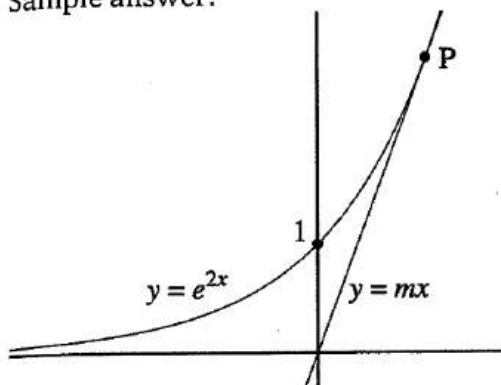
$$\begin{aligned} y &= \frac{x^3}{x+1} \\ y' &= \frac{(x+1) \times 3x^2 - x^3 \times 1}{(x+1)^2} \\ &= \frac{(2x+3)x^2}{(x+1)^2} \end{aligned}$$

34. The line $y = mx$ is a tangent to the curve $y = e^{2x}$ at a point P .

- Sketch the line and the curve on one diagram.
- Find the coordinates of P .
- Find the value of m .

34(a)

Sample answer:



34(b)

At P the y -values of the line and the curve coincide:

$$y = mx = e^{2x} \quad (1)$$

At P the slopes of the line and the tangent to the curve coincide:

$$m = \frac{d}{dx} e^{2x} = 2e^{2x} \quad (2)$$

Substitute (1) into (2):

$$m = 2mx$$

$$x = \frac{1}{2}$$

$$y = e^{2 \times 0.5} = e$$

$\therefore$ Coordinates of P are $(0.5, e)$.

34(c)

$$\text{At } P \quad mx = e^{2x}$$

$$m \times 0.5 = e^{2 \times 0.5}$$

$$m = 2e$$



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35. What is the equation of the tangent to the curve $y = (2x + 1)^4$ at the point $x = -1$?

$$y = (2x + 1)^4$$

$$y' = 4(2x + 1)^3 \times 2 = 8(2x + 1)^3$$

At the point $x = -1$:

$$y' = 8(2 \times (-1) + 1)^3 = -8$$

Equation of the tangent:

$$y - y_1 = m(x - x_1)$$

$$y - 1 = -8(x - (-1))$$

$$y + 1 = -8x - 8$$

$$8x + y + 9 = 0$$

36. What is the derivative of $\frac{x}{\cos x}$?

A. $\frac{\cos x + x \sin x}{\cos^2 x}$

B. $\frac{\cos x - x \sin x}{\cos^2 x}$

C. $\frac{x \sin x - \cos x}{\cos^2 x}$

D. $\frac{-x \sin x - \cos x}{\cos^2 x}$

36 (A)

$$\begin{aligned} \frac{d}{dx} \left(\frac{x}{\cos x} \right) &= \frac{\cos x \times 1 - x \times (-\sin x)}{\cos^2 x} \\ &= \frac{\cos x + x \sin x}{\cos^2 x} \end{aligned}$$

37. Differentiate $(\sin x - 1)^8$.

$$\frac{d}{dx} (\sin x - 1)^8 = 8(\sin x - 1)^7 \cos x$$

38. Differentiate $x^2 e^x$.

$$\begin{aligned} \frac{d}{dx} (x^2 e^x) &= x^2 e^x + e^x 2x \\ &= x e^x (x + 2) \end{aligned}$$

39. Find the equation of the tangent to the curve $y = x^2 - 5x$ at the point $(1, -4)$.

39 (A)

$$y = x^2 - 5x$$

$$y' = 2x - 5$$

At the point $(1, -4)$:

$$y' = 2 \times 1 - 5 = -3$$

Equation of the tangent:

$$y - y_1 = m(x - x_1)$$

$$y - (-4) = -3(x - 1)$$

$$y + 4 = -3x + 3$$

$$y = -3x - 1$$